

Improved Euler's method

Basic skills

Q1. $x_0 = 5$ $h = 0.2$

a) $x_1 = x_0 + h = 5 + 0.2 = 5.2$

b) $x_5 = 6$

c) $x_7 = 6.4$

d) $x_{10} = 7$

Q2. $f(x, y) = x^2 + xy$

a) $f(1, 1) = 1^2 + 1 \times 1 = 2$

b) $f(2, 3) = 2^2 + 2 \times 3 = 10$

c) $f(5, 2) = 5^2 + 5 \times 2 = 35$

d) $f(7, -4) = 7^2 + 7 \times -4 = 21$

Q3. $f(x, y) = \ln(x+y)$ $h = 0.25$

a) $x_0 = 0$, $y_0 = 1$

$$y_1^* = y_0 + h f(x_0, y_0)$$

'Old' Euler step

$$\text{so } y_1^* = 1 + 0.25 \ln(0+1) \\ = 1$$

$$\text{b) } x_0 = 3, y_0 = 1$$

$$y_1^* = 1 + 0.25 \ln(3+1) \\ = 1.346$$

$$\text{Q4. } f(x, y) = x^2 + y^2 \quad x_0 = 2, y_0 = 5 \\ h = 0.25$$

$$y_1^* = 12.25 \quad \text{and } x_1 = 2.25$$

$$y_1 = y_0 + \frac{h}{2} (f(x_0, y_0) + f(x_1, y_1^*)) \\ = 5 + \frac{0.25}{2} ((2^2 + 5^2) + (2.25^2 + 12.25^2)) \\ = 28.02$$

Competency skills

$$\text{Q5. } \frac{dy}{dx} = \overbrace{xy + x}^{f(x, y)} \quad x_0 = 0, y_0 = 1, h = 0.25$$

$$\text{a) } y_1^* = y_0 + h f(x_0, y_0) = 1 + 0.25(0 \times 1 + 0) \\ = 1$$

b) $x_1 = 0.25$

$$y_1 = y_0 + \frac{h}{2} (f(x_0, y_0) + f(x_1, y_1^*))$$

$$= 1 + \frac{0.25}{2} ((0 \times 1 + 0) + (0.25 \times 1 + 0.25))$$

$$= 1.0625$$

c) $x_2 = 0.5$

$$y_2^* = y_1 + h f(x_1, y_1)$$

$$= 1.0625 + 0.25 (0.25 \times 1.0625 + 0.25)$$

$$= 1.1914$$

$$y_2 = y_1 + \frac{h}{2} (f(x_1, y_1) + f(x_2, y_2^*))$$

$$= 1.0625 + \frac{0.25}{2} (0.25 \times 1.0625 + 0.25 + 0.5 \times 1.1914 + 0.5)$$

$$= 1.2639$$

$x_3 = 0.75$

$$y_3^* = 1.2639 + 0.25 (0.5 \times 1.2639 + 0.5)$$

$$= 1.5469$$

$$y_3 = 1.2639 + 0.125(0.5 \times 1.2639 + 0.5 + 0.75 \times 1.5469 + 0.75)$$

$$= 1.6442$$

$$x_4 = 1$$

$$y_4^* = 1.6442 + 0.25(0.75 \times 1.6442 + 0.75)$$

$$= 2.1400$$

$$y_4 = 1.6442 + 0.125(0.75 \times 1.6442 + 0.75 + 1 \times 2.1400 + 1)$$

$$= 2.2846$$

d) As $y(1) = y(x_4) = y_4 = 2.2846$

Q6. $\frac{dy}{dx} = e^{x^2}y + 3x + y$ $x_0 = 0$ $y_0 = 3$
 $h = 0.5$

$$x_1 = 0.5$$

$$y_1^* = 3 + 0.5(e^{0^2 \times 3} + 3 \times 0 + 3)$$

$$= 5$$

$$y_1 = 3 + 0.25(e^{0^2 \times 3} + 3 \times 0 + 3 + e^{0.5^2 \times 5} + 3 \times 0.5 + 5)$$

$$= 6.50$$

$$x_2 = 1$$

$$y_2^* = 6.5 + 0.5 (e^{0.5^2 \times 6.5} + 3 \times 0.5 + 6.5) \\ = 13.04$$

$$y_2 = 6.5 + 0.25 (e^{0.5^2 \times 6.5} + 3 \times 0.5 + 6.5 \\ + e^{1^2 \times 13.04} + 3 \times 1 + 13.04) \\ = 115131$$

$$\text{so } y(1) = y_2 = 115,131$$

$$\text{Q7. } \frac{dy}{dx} = (x+y)^3 \quad x_0 = 1, y_0 = 5 \\ h = 1$$

$$x_1 = 2$$

$$y_1^* = 5 + 1 ((1+5)^3) = 221$$

$$y_1 = 5 + 0.5 ((1+5)^3 + (2+221)^3) \\ = 5,544,896.5$$

$y(2)$ by Euler is 221

$y(2)$ by Improved Euler is 5.5×10^6

The reason is the gradient dramatically increases with y (and x)

Advanced skills

Q8. a) $\frac{dy}{dx} = xy + x$ ↖ could do

$$\frac{dy}{dx} = x(y+1)$$

↘ separate

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Int factor

$$I = e^{\int P(x) dx}$$

$$\int \frac{1}{y+1} dy = \int x dx$$

$$\ln(y+1) = \frac{1}{2}x^2 + C$$

$$y+1 = e^{\frac{1}{2}x^2 + C} = Ae^{\frac{1}{2}x^2}$$

$$y = Ae^{\frac{1}{2}x^2} - 1$$

b) Apply initial condition (as in Q5)

$$y(0) = 1$$

$$\text{so } 1 = A - 1 \quad A = 2$$

$$y(x) = 2e^{\frac{1}{2}x^2} - 1$$

$$y(1) = 2e^{\frac{1}{2}} - 1 \approx 2.297$$

in Qs we got 2.2846

so good approximation

c) $x_0 = 0$ $y_0 = 1$

$$x_1 = 0.25$$

Euler's (non-improved)
method

$$y_1 = 1 + 0.25(0 \times 1 + 0) = 1$$

$$x_2 = 0.5$$

$$y_2 = 1 + 0.25(0.25 \times 1 + 0.25) = 1.125$$

$$x_3 = 0.75$$

$$y_3 = 1.125 + 0.25(0.5 \times 1.125 + 0.5) \\ = 1.391$$

$$x_4 = 1$$

$$y_4 = 1.391 + 0.25(0.75 \times 1.391 + 0.75) \\ = 1.839$$

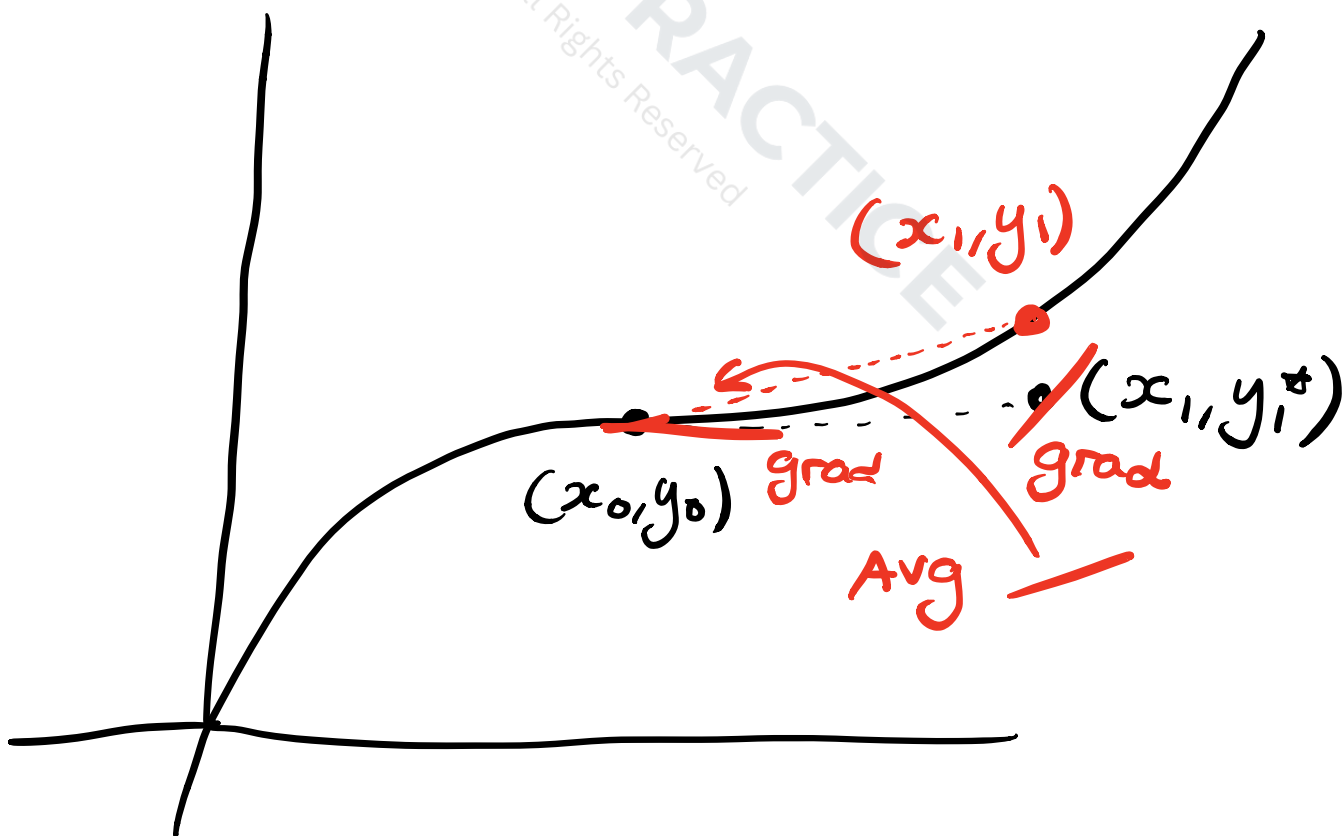
So by Euler's method get

$$y(1) = 1.839$$

not as good as improved as
not increasing quickly enough

d) It is definitely worth using the improved version.

Q9. Improved uses Average gradient between points to improve estimate



Q10.

$$\frac{d^2 y}{dx^2} = 2 \frac{dy}{dx} - 3y + x$$

$$\text{let } \underbrace{z = \frac{dy}{dx}}_{g(z)} \Rightarrow \frac{dz}{dx} = \underbrace{2z - 3y + x}_{f(x,y,z)}$$

$$x_0 = 0, y_0 = 0, z_0 = 2 \text{ with } h = 1$$

$$x_1 = 1$$

$$z_1^* = 2 + 1 \left(\underbrace{2 \times 2 - 3 \times 0 + 0}_{f(x_0, y_0, z_0)} \right) = 6$$

$$y_1^* = 0 + 1 \left(\underbrace{2}_{g(z_0)} \right) = 2$$

$$z_1 = 2 + 0.5 \left(\underbrace{2 \times 2 - 3 \times 0 + 0}_{f(x_0, y_0, z_0)} + \underbrace{2 \times 6 - 3 \times 2 + 1}_{f(x_1, y_1^*, z_1^*)} \right) = 7.5$$

$$y_1 = 0 + 0.5 \left(\underbrace{2 - 6}_{g(z_0) + g(z_1^*)} \right) = -2$$

Could use $g(z_0) + g(z_1)$

$$\text{so } y(1) = -2$$